Baseline-Relative Predictability Horizons for Cross-Domain Forecast Evaluation

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Abstract

Forecasting studies are typically summarized at a few fixed lead times, which can obscure the practical question of how long a method remains useful relative to an explicit baseline. We recast predictability-horizon estimation as a forecast evaluation task and measure baseline-relative performance through the forecast skill curve $\mathbf{Skill}(\boldsymbol{h})$, obtained using a leakage-free rolling-origin procedure with persistence as the reference. To distinguish sporadic from sustained benefits, we report two complementary horizon descriptors: $\boldsymbol{H}^*\mathbf{(relax)}$, the maximum evaluated horizon such that $\mathbf{Skill}(\boldsymbol{h}) > \mathbf{0}$ (allowing intermediate non-positive gaps), and $\boldsymbol{H}^*\mathbf{(strict)}$, the length of the longest contiguous interval of positive skill. When needed (especially for fragmented or late-recovery profiles), we additionally report interval locations and sign-change structures. A pre-specified seasonal-persistence sensitivity is evaluated on exact common support. It changes strict continuity and interval location in several domains even when relaxed reach is unchanged, confirming that the descriptors are baseline-conditional. Across the four settings, the empirical results reveal distinct horizon-wise skill-profile patterns: immediate skill collapse in electric load, delayed onset in $\mathrm{PM}_{\mathbf{2.5}}$, sustained positive skill in wind speed, and fragmented positive intervals in traffic speed. These contrasting patterns motivate reporting multiple operational descriptors beyond fixed-horizon error summaries so that total positive reach, sustained contiguous usefulness, and fragmented interval structure can be distinguished under a shared leakage-free protocol.

Keywords: forecast evaluation; predictability horizon; baseline-relative skill; rolling-origin evaluation; cross-domain forecasting

1 Introduction

In applied forecasting, evaluation is often reported at a small number of fixed lead times. This practice can obscure when (and for how long) a method is operationally useful relative to a strong baseline, especially when multi-step performance is non-monotonic or fragmented across the horizon. Methodological audits also document recurrent evaluation failures—including temporally invalid partitioning, leakage-prone preprocessing, omitted parsimonious baselines, and uncritical metric selection—that can materially distort conclusions about practical forecasting value (Hewamalage et al. 2023). Yet cross-domain studies that quantify baseline-relative usefulness over the full forecast horizon under a shared, leakage-free protocol remain comparatively uncommon.

We address this gap with leakage-free rolling-origin evaluation, train-only preprocessing, and persistence as an explicit reference baseline. Horizon-wise baseline-relative performance is reported as skill curves $\mathrm{Skill}(h)$, and operational reach is summarized using complementary horizon descriptors that separate total positive reach from sustained contiguous usefulness.

We frame predictability-horizon estimation as an evaluation-methodology contribution for forecasting systems in machine learning and data mining. The goal is not to create another applied leader board, but to establish a protocol for characterizing and comparing useful predictive reach by horizon relative to a clearly specified baseline, thereby complementing recent methodological work on multi-step forecasting (Green et al. 2025; Kreuzer et al. 2025).

Empirically, we apply this unified evaluation design to four domains—$\mathrm{PM}_{2.5}$ air quality, electric load, wind speed, and traffic speed—chosen to span heterogeneous dynamics and baseline competitiveness under the same leakage-free protocol.

The paper makes three contributions:

1. **A reproducible leakage-free evaluation framework:** we specify a rolling-origin protocol with train-only preprocessing that computes horizon-wise skill $\mathrm{Skill}(h)$ relative to an explicit persistence baseline, enabling comparable operational assessments across domains.
2. **Operational horizon descriptors for useful predictive reach and continuity:** we instantiate the generic predictability-horizon concept via two complementary descriptors—$H^*(\mathrm{relax})$, the maximum evaluated horizon such that $\mathrm{Skill}(h) > 0$ (allowing intermediate non-positive gaps), and $H^*(\mathrm{strict})$, the length of the longest contiguous positive-skill interval $[h_{\mathrm{start}}, h_{\mathrm{end}}]$—augmented, when needed, with interval locations and sign-change structure for fragmented or late-recovery profiles.
3. **A cross-domain empirical analysis under a shared protocol:** we show that domains differ not only in positive reach but also in continuity and fragmentation, and we quantify how the descriptors change under a pre-specified alternative baseline on exact common support.

Together, these contributions provide a leakage-free, baseline-relative basis for estimating and comparing operational predictability horizons across domains.

2 Related Work

2.1 Forecast evaluation metrics

Within the forecasting literature, forecast evaluation is most often framed in terms of accuracy metrics such as MAE and RMSE, along with scaled or relative variants that enable comparisons across different series and applications. Hyndman and Koehler (2006) survey the most commonly used measures and underscore that no single metric is universally ideal; the appropriate choice depends on scale, loss asymmetry, and the decisions the forecasts are intended to inform (Petropoulos et al. 2022).

For probabilistic forecasts, evaluation is commonly grounded in strictly proper scoring rules, which ensure that the expected score is optimized by the forecaster's true predictive distribution (Gneiting and Raftery 2007). This framework shifts the evaluation focus from point accuracy to the full predictive distribution, but the predominant empirical convention remains point-forecast-based reporting at fixed horizons.

When comparing forecasting methods, statistical tests of predictive accuracy (e.g., the Diebold–Mariano framework) are also frequently used to assess whether observed differences are larger than would be expected by chance (Diebold and Mariano 1995). This work does not employ formal significance tests; instead, skill is assessed relative to explicit baselines across the full set of evaluated horizons, with baseline sensitivity evaluated on exact common support.

Empirical work typically reports errors for one or a few fixed horizons. While convenient, this fixed-horizon focus can hide how performance changes with lead time, including non-monotonic, fragmented, or delayed skill profiles in multi-step settings. Operationally, it can therefore be unclear whether a method is unhelpful overall or whether it provides baseline-relative usefulness only over specific horizon intervals. Fixed-horizon reporting also makes it difficult to directly identify where horizon-wise skill becomes non-positive relative to an explicit baseline.

Large-scale benchmark exercises have reinforced this convention. Comparisons such as the M4 competition (Makridakis et al. 2020) and repositories like the Monash Time Series Forecasting Archive (Godahewa et al. 2021) assess many methods over many series, but usually at fixed or pre-chosen horizons. This setup is effective for contrasting methods on relative accuracy; yet, it does not directly characterize how forecast skill evolves over the full horizon range or at which point baseline-relative skill is exhausted.

A further drawback of fixed-horizon summaries is that, even when a strong baseline is available, they only indirectly address operational relevance. From an applied standpoint, the key question is often not whether a method lowers error at a single, predefined horizon, but whether it consistently improves on a baseline across the set of horizons that matter for planning, control, or service-level agreements. Hyndman and Athanasopoulos (2018) consolidates prevailing guidance on forecast evaluation and highlights the need for out-of-sample validation and for matching the evaluation design to how forecasts are generated and used. This points to evaluation constructs that summarize performance over horizons and that are interpretable relative to a clearly specified reference forecast. Related methodological audits emphasize that common evaluation pitfalls (e.g., leakage-prone splits and missing strong baselines) can materially distort conclusions about practical forecasting value (Hewamalage et al. 2023); we use this as motivation for the leakage-free, baseline-relative protocol described in Section 4.

2.2 Forecast skill evaluation

Baseline-relative evaluation is often expressed in terms of forecast skill, which measures the error of a candidate forecasting method against that of a reference forecast:

$$\mathrm{Skill}(h) = 1 - \frac{E_{\mathrm{model}}(h)}{E_{\mathrm{baseline}}(h)}. \tag{1}$$

This representation is fundamental in forecast verification, where the quality of a forecast is defined only in relation to a baseline (Murphy 1993). In practical forecasting settings, the same rationale underpins operational interpretation: a model that marginally improves absolute error may still be operationally unimportant if it does not surpass a strong baseline at the horizons that matter, whereas a model whose gains appear only after a period of short-horizon baseline dominance can still be highly useful for longer-range decision-making.

Skill-based reporting aligns naturally with multi-horizon evaluation because it produces a horizon-specific quantity, the skill curve $\mathrm{Skill}(h)$, instead of a single aggregate metric. This facilitates distinguishing between evaluation outcomes driven by the strength of the baseline and those stemming from modeling decisions. It also underscores the importance of rigorous temporal evaluation protocols. Out-of-sample forecast tests are typically implemented via time-aware designs such as rolling-origin evaluation, which repeatedly re-fits models and scores forecasts as the forecast origin moves forward in time (Tashman 2000). These designs mirror real forecasting practice and mitigate the risk that conclusions hinge on a particular train–test split. The suitability of rolling-origin designs, in contrast to blocked or random cross-validation for time series, has been investigated empirically, with evidence indicating that respecting temporal order is crucial for reliable performance estimation (Cerqueira et al. 2020; Bergmeir and Benítez 2012).

In machine-learning-based forecasting, this principle is coupled with a further practical issue: preventing temporal leakage arising from preprocessing, feature engineering, hyperparameter tuning, or validation procedures that inadvertently exploit future information (Kapoor and Narayanan 2023). Such leakage can artificially boost apparent skill, especially at short horizons where persistence effects are strong and where small misalignments or preprocessing-related leakage can dominate evaluation results. Therefore, leakage-free temporal validation is essential for interpreting skill curves and for using baseline-relative evaluation to draw operational conclusions about when forecasts are genuinely beneficial.

2.3 Predictability limits

Finite predictability in complex systems has long been recognized as a core issue in forecasting and dynamical systems. Lorenz (1969) demonstrated that sensitivity to initial conditions can create inherent constraints on the lead times over which accurate prediction is achievable. In empirical forecasting, similar limitations appear as the horizon grows and the baseline becomes increasingly difficult to surpass. However, these limits are not solely characteristics of a particular model class; they emerge from the interplay among the system's dynamics, the choice of reference forecast, and the evaluation protocol.

Fixed-horizon accuracy metrics remain the dominant reporting convention (Makridakis et al. 2020; Godahewa et al. 2021), but they do not directly identify the horizon at which baseline-relative skill is exhausted. From an operational standpoint, the key quantity is the range of horizons over which a method achieves positive skill relative to a baseline, under a clearly specified, temporally coherent evaluation design.

A further limitation is comparability across heterogeneous domains. Although baseline-relative skill measures are well established (Hyndman and Koehler 2006; Murphy 1993; Tashman 2000), the literature less frequently provides cross-domain descriptions of practically useful predictability horizons under a shared, leakage-free protocol. Large benchmark studies (Makridakis et al. 2020; Godahewa et al. 2021) compare methods based on accuracy, but typically do not interpret the results as operational estimates of when baseline-relative skill is depleted.

Without such a shared protocol, it remains unclear whether differences in reported predictability across domains reflect genuine differences in system dynamics and baseline competitiveness, or merely reflect inconsistencies in evaluation design, preprocessing choices, or baseline specification. This gap motivates baseline-relative horizon descriptors that summarize the span of positive skill in a way that is robust to fragmented or slowly emerging skill profiles and that supports comparison across forecasting systems without conflating predictability with methodological novelty or dataset-specific tuning.

3 Problem Formulation

Consider a time series y_t, $t = 1, \ldots, T$. A forecasting model generates predictions

$$\hat{y}_{t+h} = f_\theta(y_t, \ldots, y_{t-L+1}), \tag{2}$$

where L denotes the input window length and h the forecast horizon. The corresponding forecast error is

$$e_{t,h} = y_{t+h} - \hat{y}_{t+h}. \tag{3}$$

An aggregated error measure at horizon h is given by $E(h)$, for example, MAE or RMSE. A baseline forecasting method $b(\cdot)$ induces a baseline error curve $E_{\text{baseline}}(h)$.

3.1 Operational Predictability Horizon

In this work, operational predictability is understood as the range of time over which a forecasting system maintains positive skill relative to an explicit baseline, under a fixed, leakage-free evaluation setup.

We use H^* as the generic conceptual notation for the operational predictability horizon: the maximum forecast horizon at which a model maintains positive skill relative to an explicit baseline under a specified evaluation protocol. Formally, the generic *Forecast Skill Horizon* is

$$H^* = \max\{h \in \mathcal{H} : \mathrm{Skill}(h) > 0\}, \qquad \mathrm{Skill}(h) = 1 - \frac{E_{\mathrm{model}}(h)}{E_{\mathrm{baseline}}(h)}. \tag{4}$$

Here, $\mathcal{H}$ denotes the set of evaluated forecast horizons. H^* is protocol-defined: its value depends on the baseline choice (persistence in the primary analysis), preprocessing strategy, error metric, evaluated support, and evaluation scheme (leakage-free rolling-origin). This dependency is intentional, as it makes the descriptor reproducible and operationally interpretable.

Operational definitions:

In empirical multi-step settings, Skill(h) curves may be non-monotonic, fragmented, or exhibit delayed recovery, so the single generic definition above is best treated as a conceptual anchor. For example, a curve with early negative skill but a late positive recovery yields a large H^*(relax) but a short H^*(strict)—two descriptors that a single maximum-horizon summary conflates.

We therefore instantiate H^* through two complementary reported descriptors, defined over the evaluated horizon set $\mathcal{H} = \{1, 2, \ldots, H_{\max}\}$ and the positive-skill subset $\mathcal{H}^+ = \{h \in \mathcal{H} \mid \mathrm{Skill}(h) > 0\}$. In the remainder of the paper, H^* is used as the generic conceptual label for the operational predictability horizon, whereas all empirical reporting is based on the two instantiated descriptors H^*(relax) and H^*(strict).

H^*(relax)**:** The relaxed horizon is the maximum evaluated horizon such that Skill(h) > 0, allowing intermediate horizons with non-positive skill.

$$H^*(\mathrm{relax}) = \max\bigl(\mathcal{H}^+ \cup \{0\}\bigr). \tag{5}$$

Intermediate horizons with Skill(h) ≤ 0 may therefore lie inside the relaxed reach, but those intermediate horizons are not themselves interpreted as skillful horizons.

H^*(strict)**:** To capture sustained usefulness, let $\mathcal{I}$ be the collection of all contiguous integer intervals $I = [h_{\mathrm{start}}, h_{\mathrm{end}}]$ with $I \subseteq \mathcal{H}^+$. The strict horizon is the *length* of the longest such interval:

$$H^*(\mathrm{strict}) = \max_{I \in \mathcal{I}} |I|, \qquad |I| = h_{\mathrm{end}} - h_{\mathrm{start}} + 1. \tag{6}$$

We additionally report the maximizing interval location $[h_{\mathrm{start}}, h_{\mathrm{end}}]$ and, for fragmented or late-recovery profiles, the full sign-change structure of Skill(h). If several intervals tie for maximum length, the earliest interval is reported. H^*(time) denotes the wall-clock duration corresponding to H^*(strict) (not to H^*(relax)), and it is reported alongside it.

Together, these two descriptors distinguish total positive reach from sustained contiguous usefulness — a distinction that matters in the fragmented and delayed-onset domains studied below. Figure 1 illustrates this distinction: relaxed reach extends to the last positive-skill horizon even when intervening gaps occur, whereas strict reach measures the longest uninterrupted positive-skill interval.

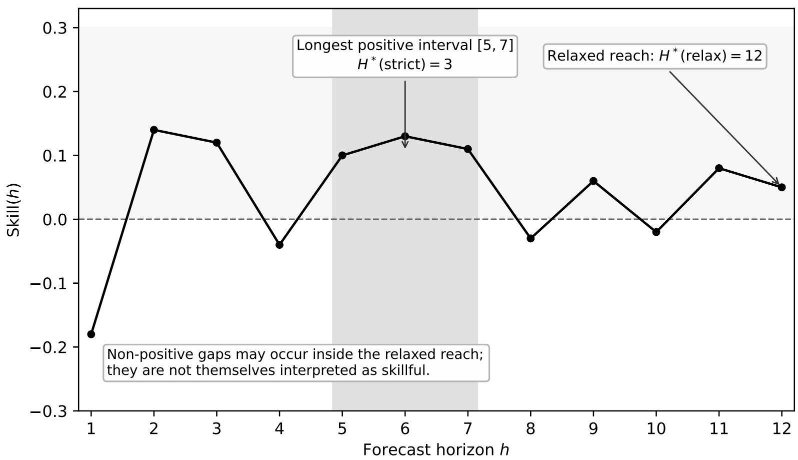

Fig. 1 Conceptual illustration of a baseline-relative skill curve Skill(h) with both positive and negative regions. The relaxed horizon H^*(relax) is the maximum evaluated horizon with Skill(h) > 0, whereas the strict horizon H^*(strict) is the length of the longest contiguous positive-skill interval $[h_{\mathrm{start}}, h_{\mathrm{end}}]$.

3.2 Operational interpretation and actionability

H^*(relax) describes the outer extent at which positive baseline-relative skill is still observed; it does not license forecasts at every preceding horizon. H^*(strict), reported together with $[h_{\mathrm{start}}, h_{\mathrm{end}}]$, identifies the longest continuous decision-relevant window in which the model improves on the chosen baseline. Fixed-horizon summaries cannot reveal delayed onset, intermediate gaps, fragmented positive islands, interval location, or late recovery. The two descriptors therefore answer different operational questions: how far any positive skill reaches, and where the longest continuous useful window lies.

4 Evaluation Framework

This section specifies the evaluation components used to compute horizon-wise baseline-relative skill curves Skill(h) and the associated horizon descriptors: the reference baseline, error metrics, and the leakage-free rolling-origin validation protocol.

4.1 Baseline models

In every domain, we evaluate models against a persistence baseline. At forecast origin t, the baseline prediction for horizon h is simply the most recent observation available at time t. Persistence is intentionally adopted as a strong operational reference because it is easy to implement, transparent, and often highly competitive for short lead times (Murphy 1993; Hyndman and Athanasopoulos 2018). Accordingly, the evaluation objective is not just to report absolute error levels but to determine the horizons at which a forecasting method yields positive skill relative to this baseline.

To assess baseline sensitivity, we additionally pre-specify seasonal persistence before examining its results:

$$\hat{y}^{\text{season}}_{t+h} = y_{t+h-k_h s}, \qquad k_h = \left\lceil \frac{h}{s} \right\rceil, \tag{7}$$

where $s = 24$ observations in the three hourly domains and $s = 7$ in daily electric load. The ceiling term selects the latest observation at the same seasonal phase as the target that is available no later than origin t, including when $h > s$. The period is fixed from sampling frequency and calendar structure, not selected from evaluation performance. For this sensitivity, LightGBM, persistence, and seasonal persistence are regenerated together, and comparisons are permitted only when forecast origin, target timestamp, horizon, and y_{true} match exactly. We report the number of records retained on this common support. At horizons that are exact multiples of s, this seasonal rule reduces to y_t and therefore coincides with ordinary persistence. Consequently, an unchanged relaxed endpoint at such a horizon is partly structural and is not interpreted as baseline invariance; changes in continuity and interval location remain informative.

4.2 Evaluation metrics

Forecast errors are evaluated primarily using mean absolute error (MAE) (Hyndman and Koehler 2006), which serves as the basis for computing baseline-relative skill at each horizon. Root mean squared error (RMSE) is additionally reported as a complementary error measure, but the main operational analysis throughout this work is framed in terms of Skill(h) computed from MAE. This design choice ensures that horizon-specific comparisons remain straightforward to interpret: at each lead time, we can directly assess whether a given forecasting approach improves upon persistence.

4.3 Temporal validation protocol

All experiments adopt rolling-origin evaluation with train-only preprocessing and persistence as the explicit reference baseline. This design yields leakage-free, out-of-sample horizon-wise skill estimates that are directly interpretable relative to a transparent, parsimonious operational baseline. Figure 2 shows how the training information set, forecast origin, and target horizons advance without exposing future observations to model fitting or preprocessing.

All experiments use strictly time-ordered, leakage-free evaluation. The procedure adopts a rolling-origin framework (Tashman 2000):

1. define a chronologically ordered training window,
2. fit any preprocessing steps using training data only,
3. produce model and persistence forecasts for horizon h,
4. advance the forecasting origin,
5. iterate over all horizons and origins.

Random train-test splits are intentionally avoided because they may introduce temporal leakage. For all domains, model and baseline forecasts are generated and compared only at aligned timestamps.

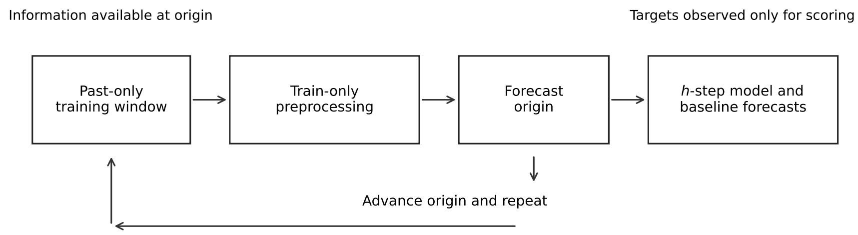

Fig. 2 Schematic of the leakage-free rolling-origin evaluation protocol. Each origin uses a past-only training window with train-only preprocessing, generates h-step-ahead forecasts for the model and the persistence baseline, and then rolls the origin forward in time.

5 Datasets

Table 1 provides an overview of the application domains, datasets, target variables, temporal resolution, and maximum forecast horizon.

The four domains are selected to support methodological comparison under a shared, leakage-free evaluation design rather than to provide exhaustive coverage of forecasting applications. They span heterogeneous temporal structures (from highly regular aggregate demand to intermittent traffic and wind dynamics), differ markedly in baseline competitiveness against persistence, and vary in temporal resolution and operational context (hourly environmental and mobility signals versus a daily energy series). Together, they form a diverse but controlled public cross-domain testbed for studying horizon-wise baseline-relative skill and its summaries.

Under this shared protocol, the four domains yield distinct baseline-relative skill-curve profiles, enabling the comparison of operational reach, contiguous usefulness, and (when present) fragmented interval structure.

Domain	Dataset	Variable	Frequency	$H_{\max}$
Air quality	UCI Beijing PM2.5 (2010–2014)	$PM_{2.5}$	hourly	48
Energy	UCI Electricity Load Diagrams 2011–2014	Aggregate load	daily	1
Wind	NREL Wind Toolkit	Wind speed	hourly	48
Traffic	PeMS / METR-LA	Traffic speed	hourly	72

Table 1 Public datasets used for cross-domain predictability analysis. In the energy domain, we use the UCI Electricity Load Diagrams 2011–2014 dataset (Trindade 2015). The aggregate daily series is obtained by summing all 370 client series and resampling to daily totals, starting from 2013-03-06, which is the first date with $\geq$90% active clients over seven consecutive days.

In the traffic domain, the raw source variable corresponds to METR-LA sensor speed measurements (Li et al. 2018).

5.1 Data representation

For each domain, the cleaned data used in the experiments is standardized into a univariate format containing a timestamp column and a single target variable. In the electric-load domain, the cleaned representation is (`timestamp`, `value`) at daily resolution. The aggregate series is defined as the daily sum of all available client-level series from the stable-coverage start date onward. This explicit specification of the aggregation procedure promotes reproducibility and ensures alignment between data preparation and experimental evaluation.

6 Experimental Setup

6.1 Forecast horizons

The maximum forecast horizons are chosen to balance practical operational relevance with a sufficient length to capture the onset and progression of skill decay:

- $PM_{2.5}$: $H_{\max} = 48$,
- Electric load (daily): $H_{\max} = 1$,
- Wind: $H_{\max} = 48$,
- Traffic: $H_{\max} = 72$ (hourly).

When supported by operational considerations, these upper limits may differ across domains. Nonetheless, the primary cross-domain analysis is carried out over a shared interval, $h = 1, \ldots, 48$, to maintain comparability among systems.

6.2 Models

The empirical study compares forecasting performance against an explicit persistence baseline and employs LightGBM as the principal forecasting model. Evaluation semantics are shared across domains, while the fixed feature and estimator configurations follow the experiment scripts that produced the submitted results.

LightGBM is a decision tree-based gradient boosting framework that is particularly well suited for tabular, lag-based forecasting formulations (Ke et al. 2017). For the three hourly domains ($PM_{2.5}$, wind, and traffic), the fixed configuration is:

- input lags: $\{0, 1, 2, 3, 6, 12, 24, 48\}$ relative to the forecasting origin,
- rolling-origin stride: 24,
- number of estimators: 50,
- learning rate: 0.05; number of leaves: 31; subsample and feature fractions: 0.9; random seed: 42.

Lag 0 (y_t) is included so that the model and baseline share the same information at the forecasting origin; this does not introduce leakage since the target remains y_{t+h} for $h \geq 1$. The daily electric-load experiment uses the fixed historical producer configuration: 14 daily lags plus day of week, a 365-day rolling training window, 100 estimators, learning rate 0.05, 15 leaves, and random seed 42. These configurations were not selected by tuning on the evaluation period.

The persistence baseline is defined as the most recent observed value available at the prediction origin.

$$\hat{y}_{t+h}^{\text{base}} = y_t, \tag{8}$$

and is applied uniformly in all domains. This baseline is deliberately strong in operational forecasting contexts and serves as the reference level against which baseline-relative usefulness is assessed. Table 2 summarizes the baseline definition and an illustrative domain-specific example. Units follow the corresponding source metadata; for electric load, the historical producer sums 15-minute kW readings, so division by four expresses the stored aggregate as kWh.

Domain	**Target unit**	**Persistence rule**	**Illustrative explanation**
$\text{PM}_{2.5}$	$\mu\text{g}\,\text{m}^{-3}$	$\hat{y}_{t+h}^{\text{base}} = y_t$	If $y_t = 42\,\mu\text{g}\,\text{m}^{-3}$, every lead uses $42\,\mu\text{g}\,\text{m}^{-3}$.
Load	Sum of 15-min kW readings	$\hat{y}_{t+h}^{\text{base}} = y_t$	The stored daily aggregate follows the producer script; dividing it by four gives the equivalent energy in kWh.
Wind	$\text{m}\,\text{s}^{-1}$ at 100 m	$\hat{y}_{t+h}^{\text{base}} = y_t$	If $y_t = 6.8\,\text{m}\,\text{s}^{-1}$, every lead uses $6.8\,\text{m}\,\text{s}^{-1}$.
Traffic	mph	$\hat{y}_{t+h}^{\text{base}} = y_t$	If $y_t = 57$ mph, every lead uses 57 mph.

Table 2 Persistence definition and verified target units in each domain. Electric-load values are retained on the exact scale produced by the historical aggregation script; the stated conversion follows the source dataset's 15-minute kW sampling.

No hyperparameter tuning is carried out in any domain. The protocol is kept deliberately simple so that observed differences in skill reflect domain characteristics and baseline competitiveness rather than model optimization. The exact fixed configuration used in each domain is reported above and retained for reproducibility; no claim of identical feature construction across hourly and daily domains is made.

6.3 Reproducibility rules

All tables and figures are generated directly from the experiments carried out under the stated evaluation protocol. To support replication, the data-processing and evaluation scripts operate on the cleaned, domain-specific datasets, with each step documented so that the reported results can be reproduced from the specified inputs.

7 $\text{PM}_{2.5}$ Domain Results

7.1 $\text{PM}_{2.5}$ experimental setting

The first domain we analyze is the Beijing $\text{PM}_{2.5}$ forecasting case (Chen 2015), which provides an initial empirical demonstration of the proposed evaluation framework. The study relies on hourly measurements, a persistence baseline, and the standard comparable LightGBM setup with $H_{\max} = 48$, using MAE-based skill for horizon-wise

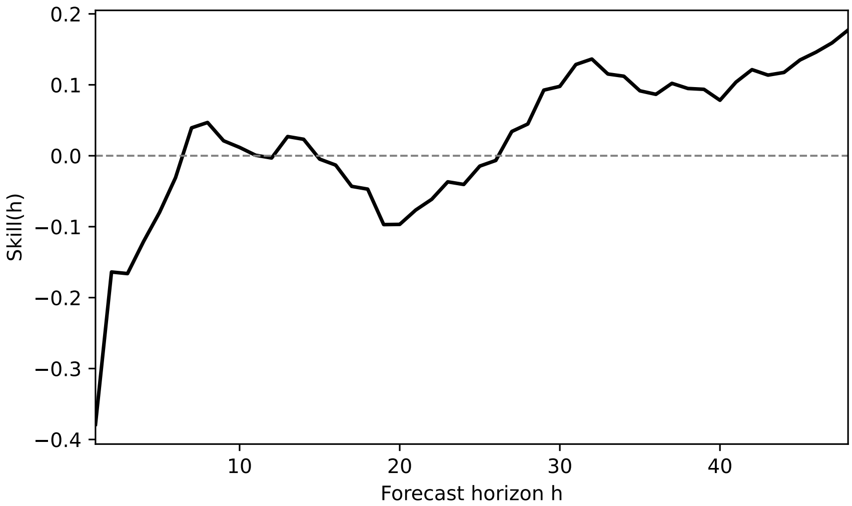

Fig. 3 $PM_{2.5}$ skill curve Skill(h) under the leakage-free rolling-origin protocol. The skill-profile pattern shows delayed onset: persistence dominates at short lead times and positive skill emerges only at longer horizons.

assessment. This domain is particularly informative for highlighting settings in which persistence remains very strong at short lead times, and positive skill appears only at longer horizons.

7.2 Observed skill behavior

For $PM_{2.5}$, the Skill(h) curve in Figure 3 shows a delayed, contiguous interval of positive skill following a long region of negative skill. Under the main protocol, this yields $H^*(\text{relax}) = 48$, $H^*(\text{strict}) = 13$, and a longest positive interval $[h_{\text{start}}, h_{\text{end}}] = [36, 48]$, with $H^*(\text{time}) = 13$ h.

7.3 Interpretation

The relaxed descriptor reaches the maximum evaluated horizon because positive skill is observed at $h = 48$, even though the early region is negative. The strict descriptor is shorter because it measures sustained usefulness and is determined by the longest contiguous positive-skill interval, here $[36, 48]$ with a length of 13. Operationally, $PM_{2.5}$ represents delayed contiguous emergence: baseline-relative usefulness appears late and persists continuously once it appears. Additional second-model artifacts for $PM_{2.5}$ (XGBoost) are available, but they are treated as qualitative support unless strict protocol comparability is explicitly documented.

8 Electric Load Domain Results

8.1 Energy experimental setting

The second domain considered is aggregate daily electricity demand, derived from the UCI Electricity Load Diagrams 2011–2014 dataset. This experiment uses daily observations, a persistence baseline, and the canonical comparable LightGBM protocol, with performance evaluated through MAE-based skill. The energy setting is especially informative because the series is highly regular, and persistence is expected to be particularly difficult to outperform; more broadly, electric load forecasting is a canonical operational setting where benchmarking and uncertainty-aware evaluation are central concerns (Hong and Fan 2016).

8.2 Observed skill behavior

In the canonical reported setting, daily electric load is operationally summarized at $h = 1$, yielding $H^*(\text{relax}) = 1$ and $H^*(\text{strict}) = 1$ with $[h_{\text{start}}, h_{\text{end}}] = [1, 1]$. For visualization purposes, Figure 4 additionally shows an auxiliary extended-horizon view up to $h = 7$ under the same protocol, highlighting that positive skill against persistence decays rapidly and remains confined to a very short early interval. Both operational variants yield the same result:

$$H^*(\text{relax}) = 1, \qquad H^*(\text{strict}) = 1, \qquad [h_{\text{start}}, h_{\text{end}}] = [1, 1], \qquad H^*(\text{time}) = 1 \text{ day.} \tag{9}$$

8.3 Interpretation

The aggregate daily electricity demand series exhibits low relative variability (CV≈0.16) and strong day-to-day autocorrelation (ACF lag-1≈0.92), both of which make the last-observation-carried-forward persistence forecast particularly difficult to improve upon at the daily scale. With only $h = 1$ evaluated, $H^*(\text{relax})$ and $H^*(\text{strict})$ coincide by construction and reflect that positive skill is observed on the first (and only) evaluated day. Operationally, this domain represents an immediate collapse under strong persistence dominance: any baseline-relative advantage is confined to the earliest evaluated step. Qualitatively similar conclusions are also supported by a simple moving-average comparison, but this is not treated as a strict protocol-matched robustness check.

9 Wind Domain Results

9.1 Wind experimental setting

The third domain considered is the NREL Wind Toolkit (Draxl et al. 2015), which provides hourly wind-speed time series. The wind experiment uses hourly observations, a persistence baseline, and the canonical comparable LightGBM protocol under $H_{\max} = 48$, with performance again evaluated through MAE-based skill. This domain is included to assess whether the proposed horizon descriptors also characterize settings in which positive skill is sustained across most or all of the evaluated horizon range.

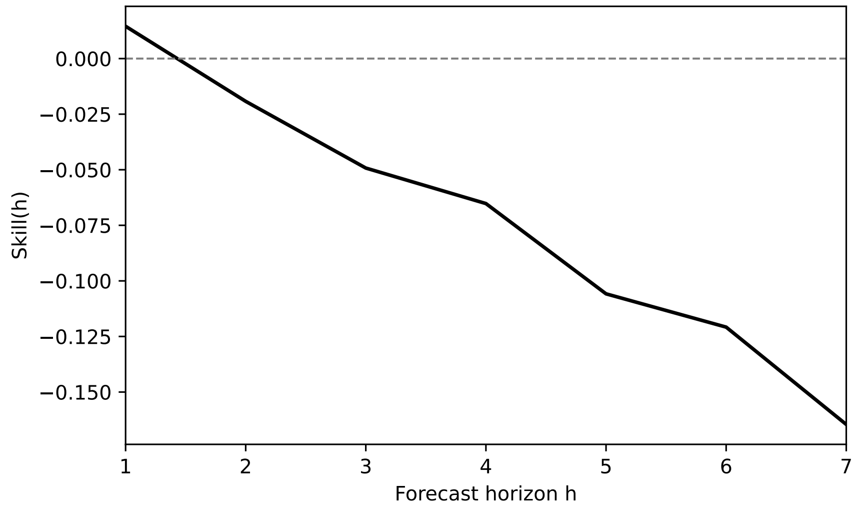

Fig. 4 Electric-load skill curve Skill(h) under the leakage-free rolling-origin protocol. The canonical cross-domain analysis uses $H_{\max} = 1$, yielding $H^*(\text{relax}) = 1$ and $H^*(\text{strict}) = 1$ at $h = 1$. The extended view up to $h = 7$ is shown only to illustrate the rapid decay of positive skill against persistence beyond the first day.

9.2 Observed skill behavior

For wind, positive skill is obtained over the entire set of evaluated horizons, yielding a uniformly positive skill curve across $h = 1, \ldots, 48$ (Figure 5). This shape indicates comparatively weak baseline competitiveness: persistence is outperformed not only at longer lead times but also at the shortest horizons. Under the operational variants examined in this study,

$$\begin{aligned} H^*(\text{relax}) &= 48, \quad [h_{\text{start}}, h_{\text{end}}] = [1, 48], \\ H^*(\text{strict}) &= 48, \qquad H^*(\text{time}) = 48 \text{ h}. \end{aligned} \tag{10}$$

9.3 Interpretation

Because the skill curve remains positive throughout the evaluated range, $H^*(\text{relax})$ and $H^*(\text{strict})$ coincide and equal $H_{\max}$, and the maximizing interval is $[1, 48]$. Operationally, wind represents sustained positive skill: the model retains baseline-relative usefulness continuously across the entire evaluated horizon range.

10 Traffic Domain Results

10.1 Traffic experimental setting

The fourth domain considered is the METR-LA hourly traffic-speed dataset (Li et al. 2018), constructed from a univariate sensor trajectory. The experiment uses hourly

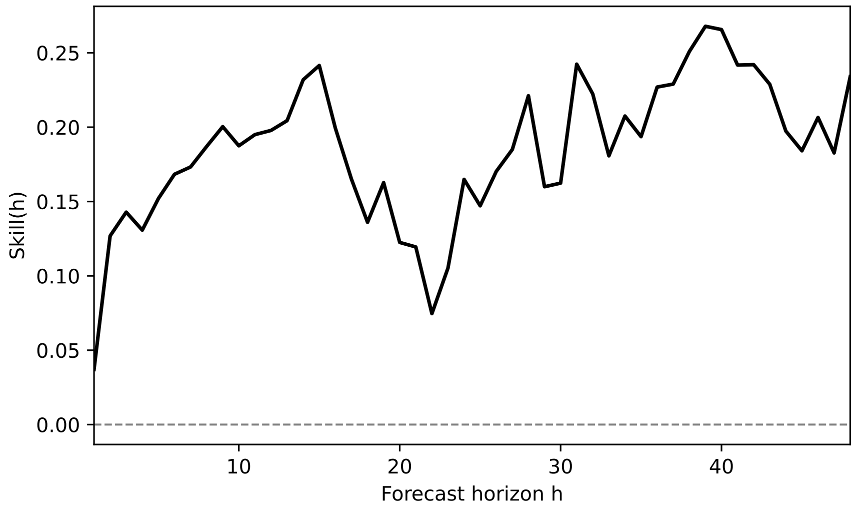

Fig. 5 Wind skill curve Skill(h) under the leakage-free rolling-origin protocol. The skill-profile pattern shows sustained positive skill across the evaluated horizon range.

observations, a persistence baseline, and the canonical comparable LightGBM protocol, with MAE-based skill evaluated for forecasting horizons up to $H_{\max} = 72$. This domain is especially suitable for studying fragmented skill profiles, in which predictive usefulness may arise in short, separated intervals rather than in a single sustained range.

10.2 Observed skill behavior

In the traffic domain, the skill curve is highly non-monotonic and fragmented under the canonical comparable LightGBM protocol (Figure 6): positive skill appears in short, separated intervals interspersed with negative regions. This shape is consistent with strong baseline competitiveness at many horizons, punctuated by intermittent recoveries where the model surpasses persistence. For LightGBM,

$$H^*(\text{relax}) = 72, \qquad [h_{\text{start}}, h_{\text{end}}] = [46, 52], \qquad H^*(\text{strict}) = 7, \qquad H^*(\text{time}) = 7 \text{ h}. \tag{11}$$

The first positive interval occurs at $[17, 18]$, and the sign of Skill(h) changes 19 times across $h \in \{1, \ldots, 72\}$.

10.3 Interpretation

These results illustrate why multiple descriptors are needed in fragmented settings. The relaxed descriptor records the last positive-skill horizon, whereas the strict descriptor and its location reveal that the longest continuous useful window is only seven hours and occurs at $[46, 52]$.

Table 3 presents the operational horizon descriptors derived for the four study domains under a common evaluation framework. Together, these four domains

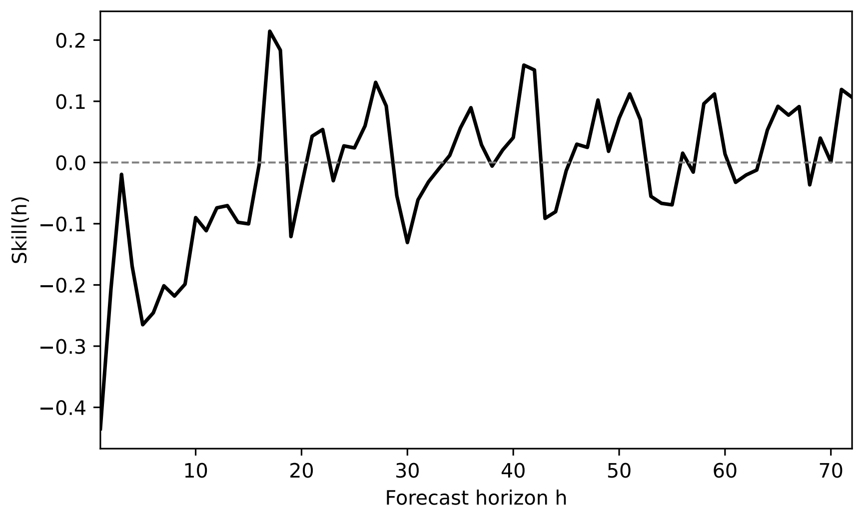

Fig. 6 Traffic skill curve Skill(h) under the leakage-free rolling-origin protocol. The skill-profile pattern shows fragmented positive intervals: brief, separated stretches of positive skill interspersed with negative regions.

instantiate qualitatively different horizon-wise skill structures under a shared, leakage-free evaluation design, which is why a single horizon summary is insufficient. The comparison reveals four distinct patterns of predictability. Electric load is characterized by immediate persistence dominance, with positive skill confined to the first day. Traffic displays fragmented predictability, where positive skill arises in short, isolated segments rather than as a continuous region. $PM_{2.5}$ exhibits a delayed onset of a contiguous interval of useful predictability, whereas wind demonstrates the highest level of operational predictability, maintaining positive skill across the entire evaluated horizon. Collectively, these findings indicate that fixed-horizon summaries can obscure substantial cross-domain variability in the structure of baseline-relative predictability.

10.4 Alternative-baseline sensitivity

Table 4 reports the pre-specified seasonal- persistence sensitivity. Every row is computed from newly generated per-origin forecasts after verifying exact equality of forecast origin, target timestamp, horizon, and y_{true}; no records are lost in the pairwise alignments. Because $PM_{2.5}$ contains missing observations, its persistence row on this common subset (48/22, $[27, 48]$) differs from the primary full-support persistence result (48/13, $[36, 48]$). This is a support effect and the sensitivity row does not replace the primary result.

Relative to the matched persistence row, seasonal persistence changes H^*(strict) from 22 to 24 in $PM_{2.5}$ and shifts the interval from $[27, 48]$ to $[25, 48]$; load remains 1/1 at $[1, 1]$; wind changes from 48 at $[1, 48]$ to 30 at $[19, 48]$; and traffic changes from 7 at $[46, 52]$ to 14 at $[48, 61]$. In all four domains H^*(relax) is unchanged. For the hourly domains, however, the evaluated endpoint is an exact multiple of the 24-hour period,

Domain	H^*(relax)	H^*(strict)	$[h_{\text{start}}, h_{\text{end}}]$	H^*(time)	Notes
$PM_{2.5}$	48	13	$[36, 48]$	13 h	strong persistence; contiguous positive-skill interval emerges late, following an extended negative region
Load (daily)	1	1	$[1, 1]$	1 day	positive skill is confined to the first evaluated day under this setup
Wind	48	48	$[1, 48]$	48 h	positive skill over the full evaluated range relative to persistence
Traffic	72	7	$[46, 52]$	7 h	fragmented profile with 19 sign changes and first positive interval $[17, 18]$

Table 3 Comparison of predictability horizons across domains ($PM_{2.5}$, electric load, wind, and traffic).

Domain	Baseline	Common n_h	H^*_{relax}	H^*_{strict}	Longest interval
$PM_{2.5}$	Persistence	328–346	48	22	$[27, 48]$
$PM_{2.5}$	Seasonal persistence	328–346	48	24	$[25, 48]$
Load	Persistence	301	1	1	$[1, 1]$
Load	Seasonal persistence	301	1	1	$[1, 1]$
Wind	Persistence	350–354	48	48	$[1, 48]$
Wind	Seasonal persistence	350–354	48	30	$[19, 48]$
Traffic	Persistence	102–108	72	7	$[46, 52]$
Traffic	Seasonal persistence	102–108	72	14	$[48, 61]$

Table 4 Baseline sensitivity on exact common support. n_h is the number of aligned forecast origins at each horizon (a range is shown when it varies with h). The persistence rows in this table are recomputed on the same support as seasonal persistence and therefore need not equal the primary full-support persistence descriptors.

where seasonal persistence equals ordinary persistence; the unchanged relaxed endpoint is therefore partly structural and is not evidence of general baseline invariance. The changes in strict continuity and interval location demonstrate the substantive baseline conditionality of the descriptors.

11 Discussion

The four empirical results reveal qualitatively distinct horizon-wise skill-profile patterns that fixed-horizon accuracy summaries would not recover. In electric load, skill collapses immediately against persistence, leaving no operationally useful horizon beyond the first day. Traffic exhibits a fragmented structure: positive skill arises in short, isolated intervals separated by non-positive regions, so that H^*(strict) is substantially shorter than H^*(relax). $PM_{2.5}$ presents a delayed-emergence pattern in which persistence dominates at short lead times, but a contiguous interval of positive skill appears and stabilizes at longer horizons. Wind stands apart as the only domain with positive skill sustained across the entire evaluated range relative to persistence.

These empirical patterns can be read as four operational profile types under baseline-relative, horizon-wise evaluation: (i) immediate collapse under strong baseline competition (electric load), (ii) delayed but contiguous emergence of positive skill ($PM_{2.5}$), (iii) sustained positive skill across the evaluated horizon range (wind), and (iv) fragmented late recovery with model-sensitive continuity (traffic). The typology makes explicit why a single horizon summary is insufficient: H^*(relax) alone can overstate operational usefulness in fragmented settings, while H^*(strict) alone can miss delayed positive reach. Reporting both descriptors—and, when needed, interval locations and sign-change structure—provides a compact summary of reach and continuity.

These results also reinforce concerns raised in prior methodological work: fixed-horizon reporting can hide delayed or fragmented usefulness, and leakage-prone evaluation or missing strong baselines can distort conclusions about practical value. The persistence-relative rolling-origin design used here addresses both issues directly.

Potential dynamic use. An operational system could use an H^* profile estimated on historical validation data to freeze a set of eligible forecast horizons for subsequent deployment, and later update that policy using only errors that have already been realised. This paper does not validate such an adaptive controller. Re-estimating H^*, selecting horizons, and evaluating that choice on the same errors would introduce adaptive selection bias and overfitting. A dynamic policy therefore requires a prospective, prequential, or nested evaluation in which horizon selection is separated from the data used to assess it.

Metric sensitivity. The qualitative morphology of Skill(h) is not uniformly invariant to the choice of error metric: while $PM_{2.5}$ and wind retain their main profile types under RMSE, electric load and traffic change materially. This reinforces the interpretation of H^* as a protocol-conditional descriptor rather than a quantity with universal cross-metric stability.

Baseline sensitivity. Seasonal persistence alters the longest continuous positive-skill window in $PM_{2.5}$, wind, and traffic, despite leaving relaxed reach unchanged in this design. Wind provides the clearest example: the strict interval contracts from the full $[1, 48]$ range under persistence to $[19, 48]$ under seasonal persistence. Traffic changes in the opposite direction, from a seven-hour to a fourteen-hour longest interval. These outcomes are reported without selecting the reference that gives a more favourable profile and show why the baseline is part of the estimand.

Scope of inference. Because baseline competitiveness and evaluation design shape Skill(h), operational predictability horizons are not intrinsic properties of a forecasting model or domain. They reflect the interaction of dataset, model, baseline choice, metric, evaluated range, and temporal validation protocol.

These findings point to two methodological implications for horizon-based forecast evaluation:

1. Predictability should be assessed relative to an explicit baseline; in cross-domain studies, using a common baseline is often essential for comparability.
2. Horizon estimation should differentiate between relaxed and contiguous predictability, particularly when empirical skill curves are delayed or fragmented.

Overall, fixed-horizon summaries lose structural information that matters operationally: fragmented or delayed profiles can yield the same fixed-horizon summary while implying very different decision windows. Reporting H^*(relax) and H^*(strict) together (and interval structure when needed) separates total positive reach from sustained usefulness while making explicit that any H^* estimate is conditional on the baseline, metric, and evaluation protocol.

12 Limitations

The horizon descriptors reported in this paper should be interpreted as *operationally conditional* summaries rather than as universal constants. Their values depend on the baseline choice, the maximum evaluated horizon $H_{\max}$, dataset and domain properties, and the evaluation design. Together, these factors determine what counts as useful baseline-relative reach under a leakage-free protocol. The empirical scope is limited to four datasets and their stated temporal periods, the fixed LightGBM configurations reported above, and the evaluated horizon caps. Results need not transfer to another dataset, target, model configuration, baseline, metric, or $H_{\max}$. The alternative-baseline sensitivity reported in the revision examines baseline conditionality but does not eliminate it. Seasonal persistence also coincides with ordinary persistence at exact multiples of the seasonal period, so unchanged terminal reach at those endpoints is not an independent invariance result. Moreover, no dynamic H^* controller is evaluated prospectively; the operational policy described in the Discussion remains a potential application requiring separate prequential or nested validation.

More formally, the horizon descriptors reported here should be understood as

$$H^* = H^*(\text{dataset}, \text{model}, \text{baseline}, \text{metric}, \text{validation protocol}),$$

rather than as intrinsic properties of a forecasting system. In particular, the metric-sensitivity analysis indicates that changing the error metric from MAE to RMSE can modify not only the numerical values of H^*(relax) and H^*(strict) but, in some domains, the qualitative morphology of Skill(h) itself. As a result, claims of cross-metric invariance are not warranted in general, and any reported horizon estimates should always be interpreted in the context of the full evaluation protocol.

These results show why H^*(relax) alone is not sufficient in fragmented settings. When skill curves are non-monotonic, total positive reach can remain stable even though contiguous usefulness and interval structure change across models. Reporting H^*(strict), together with interval locations and, when needed, sign-change structure, therefore provides a more informative summary of operational usefulness. Clear reporting of the baseline, evaluated horizon range, dataset, model, and protocol remains necessary for reproducibility and for the correct interpretation of any reported H^* estimate.

13 Conclusion

This study frames predictability-horizon estimation as an operational forecast-evaluation task, focused on skill relative to an explicit baseline. The contribution is not a new forecasting model but a leakage-free, baseline-relative way to summarize useful predictive reach across horizons via the skill curve $\mathrm{Skill}(h)$.

Using rolling-origin evaluation against persistence, we report two complementary horizon descriptors: $H^*(\mathrm{relax})$ (the maximum evaluated horizon with any positive skill) and $H^*(\mathrm{strict})$ (the longest uninterrupted run of positive skill), augmented when needed with interval locations and sign-change structure. Across the four domains, useful predictive reach differs not only in extent but also in continuity and fragmentation, and the baseline-sensitivity analysis makes explicit that these descriptors are conditional on the chosen reference. Future work will test whether these operational skill-profile types recur systematically across broader domain families and evaluation designs, and whether more expressive model classes or multivariate formulations alter the interval structure of baseline-relative usefulness.

A.1 Metric sensitivity

The main analysis in this paper is based on MAE-derived skill curves, which are a common choice for scale-dependent error evaluation in applied forecasting. To examine how the proposed horizon descriptors behave under a different loss geometry, we recomputed the skill curves using RMSE as the underlying error metric while keeping all other components of the protocol fixed (datasets, models, persistence baseline, and rolling-origin design).

The resulting metric sensitivity is heterogeneous across domains. In the electric-load domain, replacing MAE with RMSE extends the positive-skill region from the first horizon to a short early interval, yielding $H^*(\mathrm{relax}) = 3$, $H^*(\mathrm{strict}) = 3$, and $[h_{\mathrm{start}}, h_{\mathrm{end}}] = [1, 3]$ under the present configuration, without altering the broader impression that persistence dominates beyond very short lead times. In the traffic domain, the effect is considerably stronger: the MAE-based fragmented profile becomes a near-contiguous positive-skill regime under RMSE, with $H^*(\mathrm{relax}) = 72$, $H^*(\mathrm{strict}) = 63$, and the longest positive interval $[10, 72]$, indicating that the qualitative morphology of $\mathrm{Skill}(h)$ itself can depend on the error metric.

By contrast, $PM_{2.5}$ and wind retain their main qualitative profile types under RMSE. For $PM_{2.5}$, RMSE preserves the delayed-emergence pattern, with $H^*(\mathrm{relax}) = 48$, $H^*(\mathrm{strict}) = 18$, and the longest positive interval $[31, 48]$. For wind, the skill curve remains fully positive across the evaluated range, with $H^*(\mathrm{relax}) = 48$, $H^*(\mathrm{strict}) = 48$, and $[1, 48]$. Taken together, these observations suggest that the proposed descriptors should be interpreted as protocol-conditional summaries: the choice of error metric is a substantive component of the operational definition of $H^*(\mathrm{relax})$ and $H^*(\mathrm{strict})$, rather than a secondary implementation detail, and cross-metric invariance cannot be assumed uniformly across domains.

Declarations

Funding

The author did not receive support from any organization for the submitted work.

Competing Interests

The author has no competing interests to declare that are relevant to the content of this article.

Ethics Approval

Not applicable. This study is based exclusively on publicly available benchmark datasets. It did not involve human participants, animals, patient data, or primary data collection.

Data Availability

All datasets used in this study are publicly available. The replication package accompanying this revision provides a single reproducibility entry point (`data/README.md`) and a machine- readable manifest (`data/MANIFEST.tsv`) record the source identifier, expected SHA-256 checksum, preprocessing script, and redistribution status for each of the four experimental inputs. The exact processed $PM_{2.5}$ and electric-load inputs are versioned in the repository. Wind (NREL WTK-LED CONUS, 2019, point sample) and METR-LA are deterministically retrieved and processed by the supplied retrieval script, which verifies their expected checksums; they are not claimed to be repository-cached. The original sources remain the UCI repositories for electric load (Trindade 2015) and Beijing $PM_{2.5}$ (Chen 2015), NREL for wind (Draxl et al. 2015), and the DCRNN/METR-LA distribution for traffic (Li et al. 2018).

Code Availability

The replication package accompanying this revision contains the data-processing, forecasting, common-support validation, descriptor, table, and figure-generation scripts used for the reported results. The public project location is https://github.com/EduIntellect/paper2H.

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